Lab Title: Case Study 2 - Android 10 Pixel 3 Evidence Investigation

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Student ID: 2026/DFIT/17036

Course Name: Mobile and IOT Forensics (CIP-B106)

Date of Submission: 10/10/2026

Instruction Name: Aminu Idris

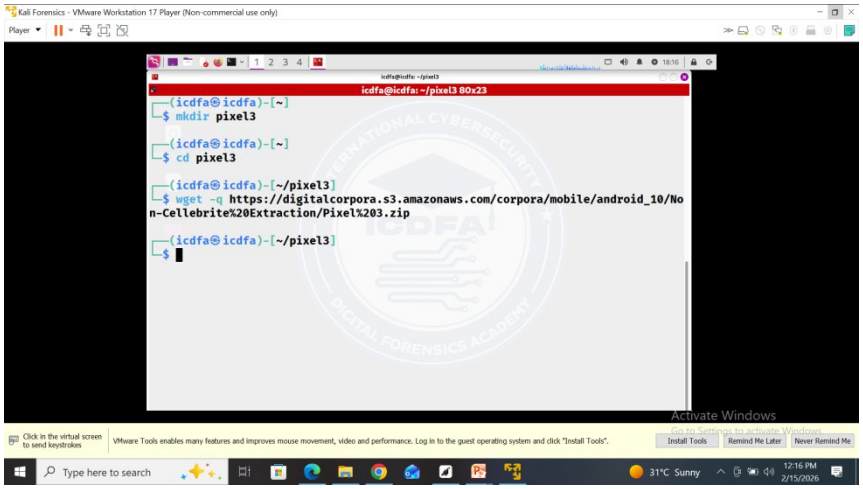
Platform Used: ICDFA Kali Linux

Executive Summary

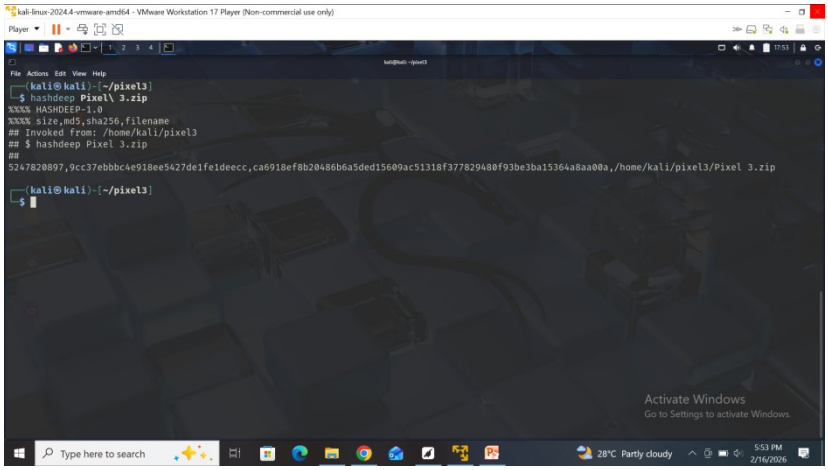
This case study examines the authorised Pixel 3 Android 10 training logical copy to identify device and account information, system settings, installed applications, communications, media, and location-related artefacts. The investigation uses system files, application databases, and usage records to reconstruct a supported chronology of relevant activity. Findings are documented and corroborated where possible, with timestamps validated and limitations clearly stated. The evidence is a logical copy acquired after TWRP root access, not a bit-for-bit physical image. Therefore, some data may be unavailable, and the acquisition process does not prove that the device user rooted the phone. All conclusions are limited to the available and verifiable evidence.

Pixel 3 Image Description

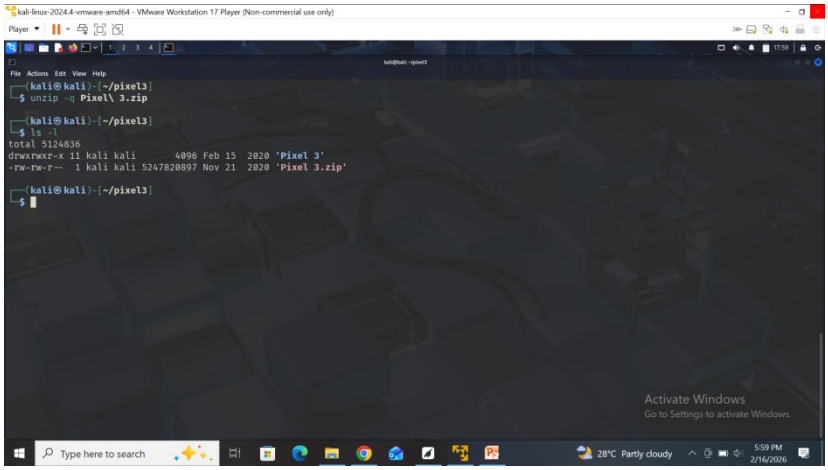
1. I downloaded Pixel 3



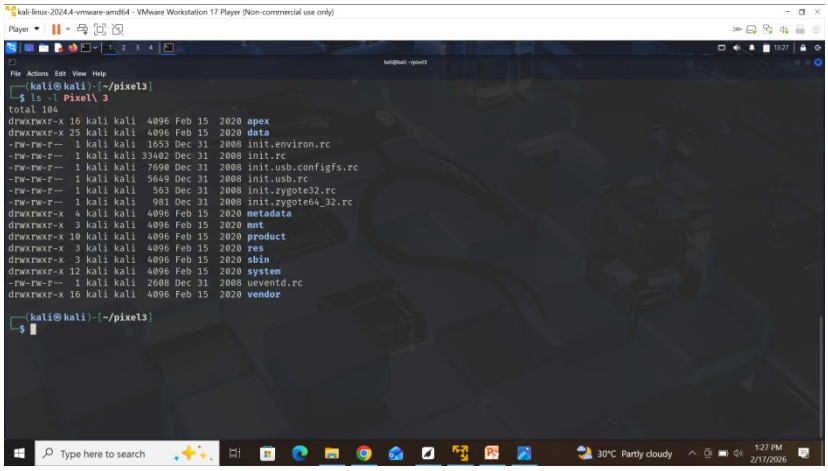
2. Verify hashes



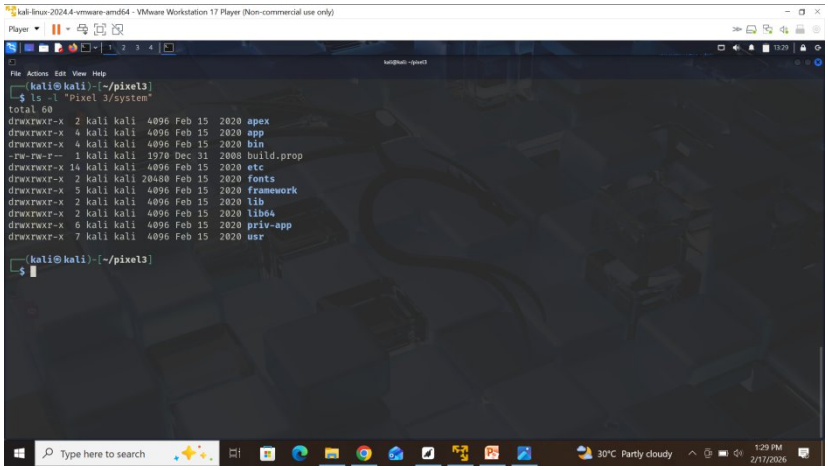
3. Unzip to get Pixel 3 image



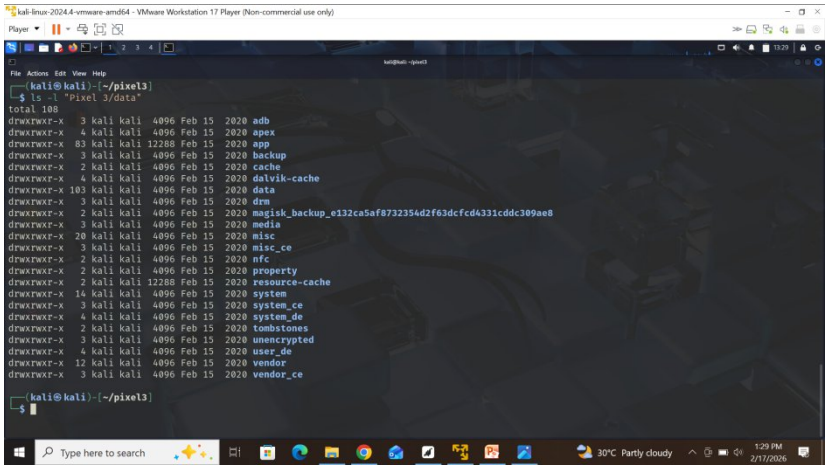
4. I listed the root directory here



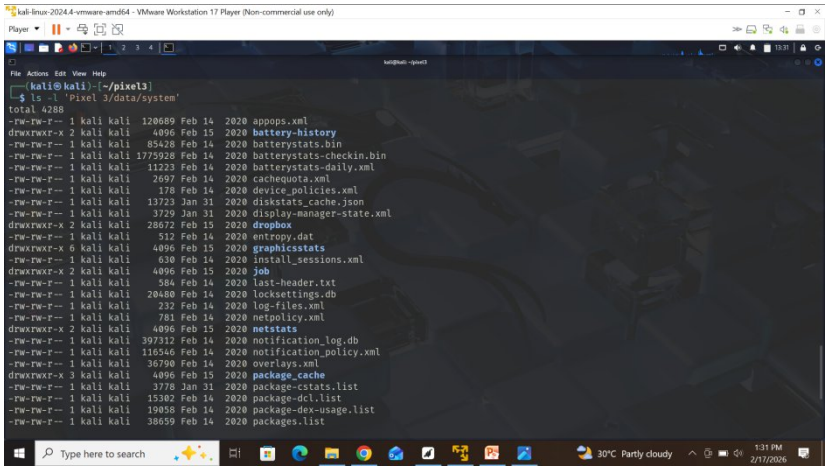
5. /System: Android OS Project (AOSP)



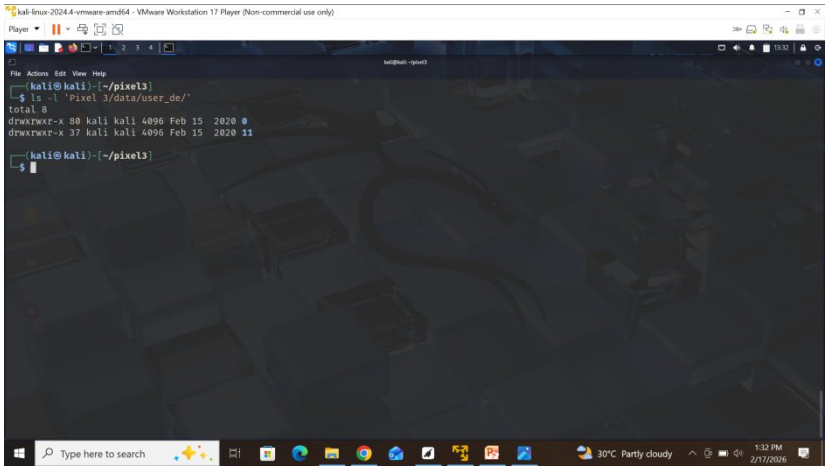
6. /data Contains subfolders with forensic data



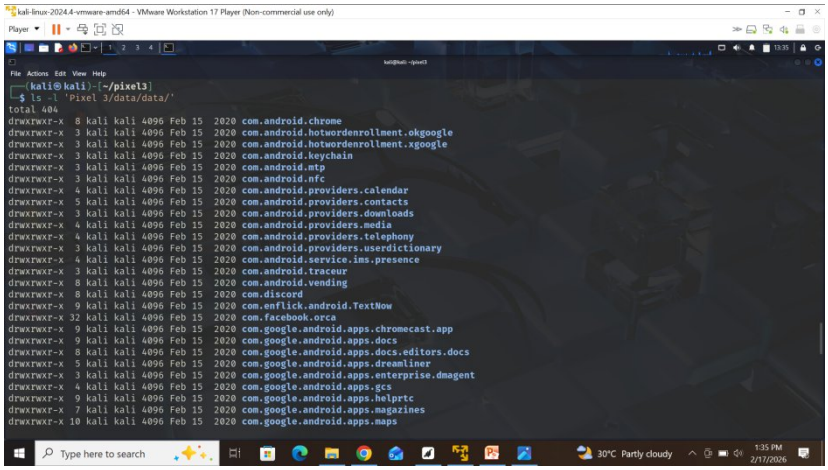
7. *data/system settings*



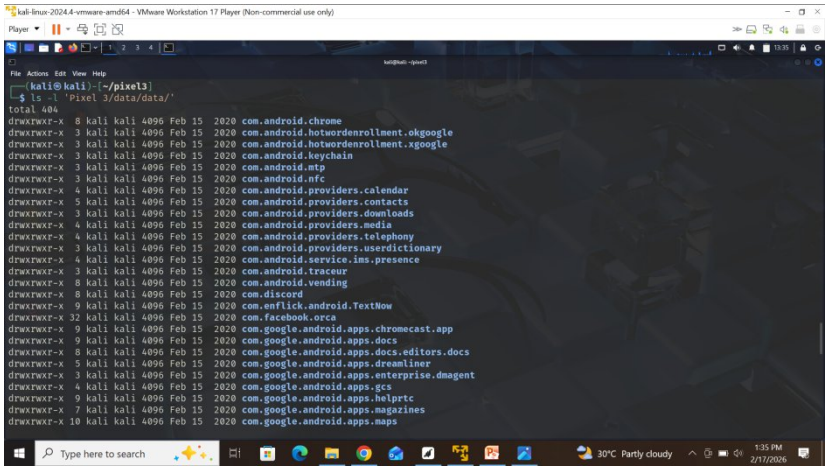
8. /data/user_de/



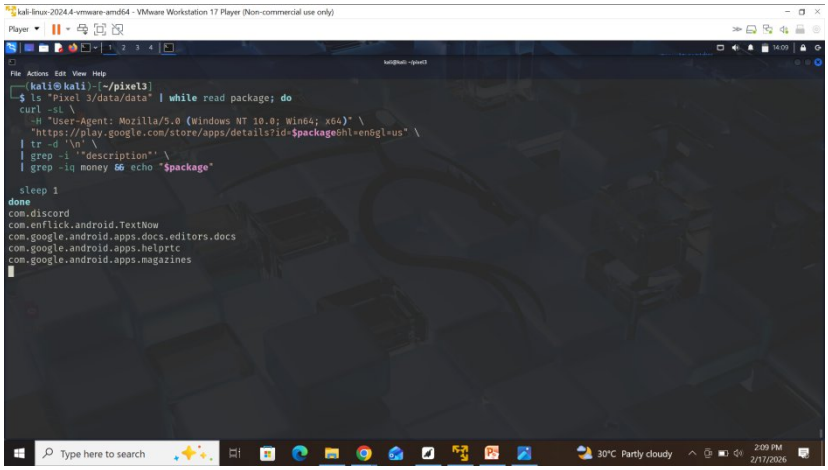
9. /data/data/



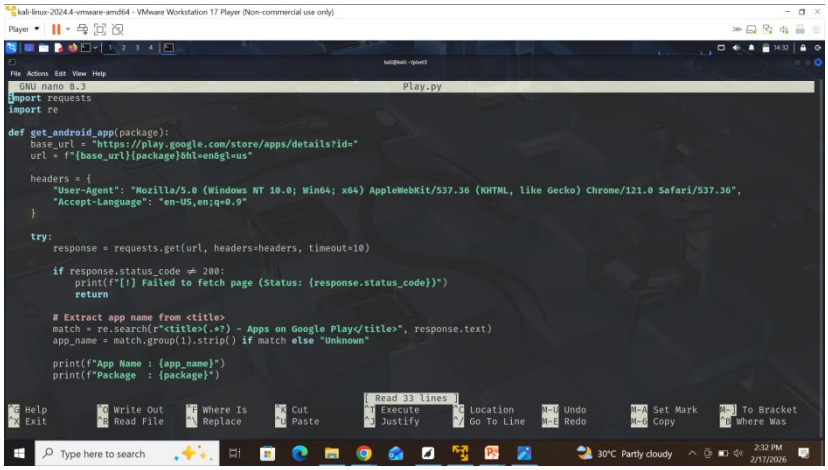
10. Find inspect the Twitter name location in Chrome

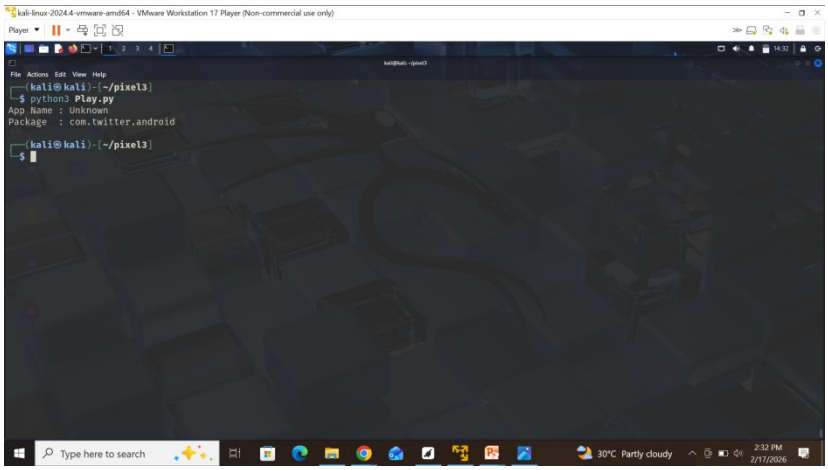


11. List all app names and package ID



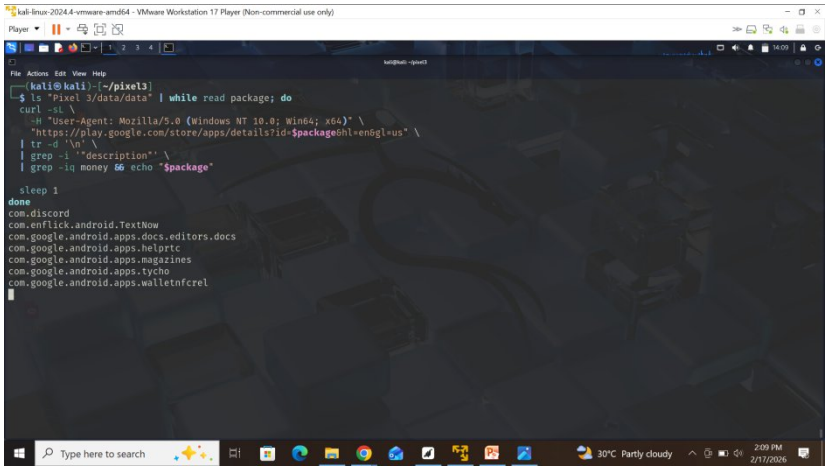
12. List all app names and package ID/ works for new Playstore HTTP response (Python version)



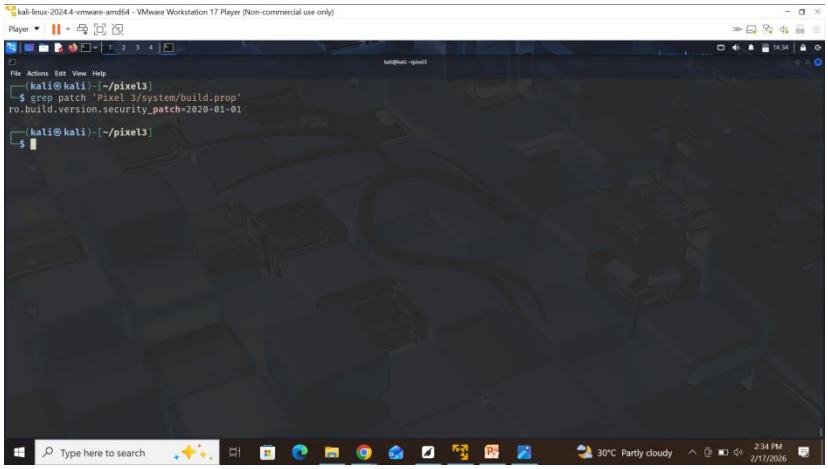


Pixel3 Device Investigations

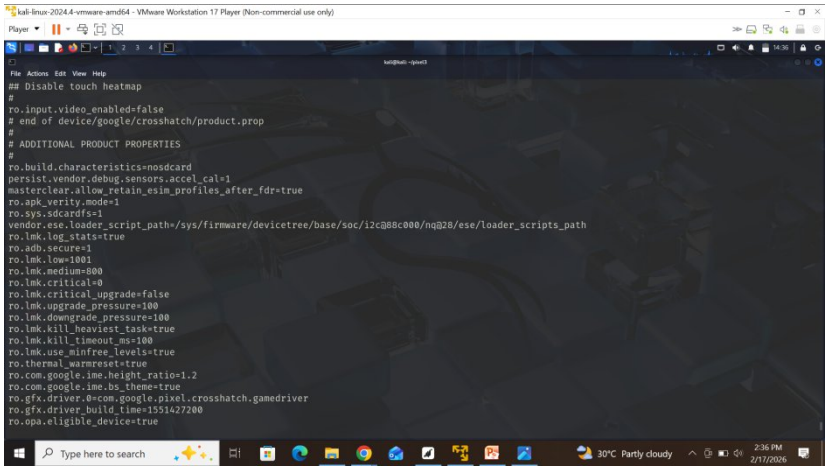
1. What is the product (phone) information?



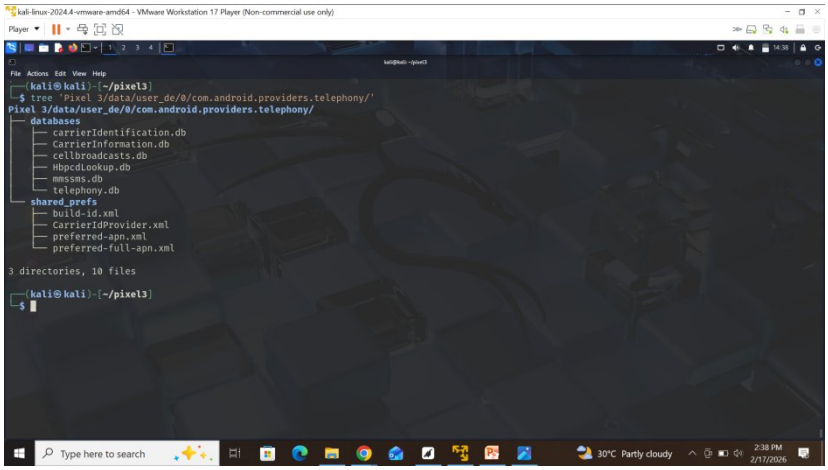
2. What is the latest security patch?



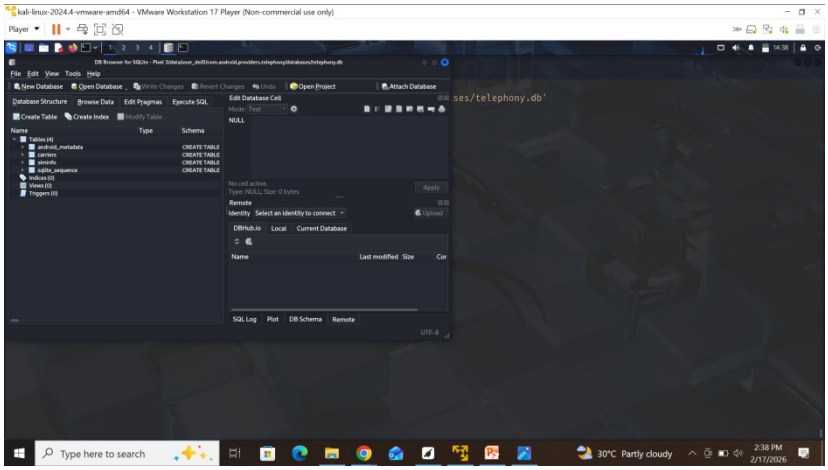
3. Does the Pixel have a SD card?

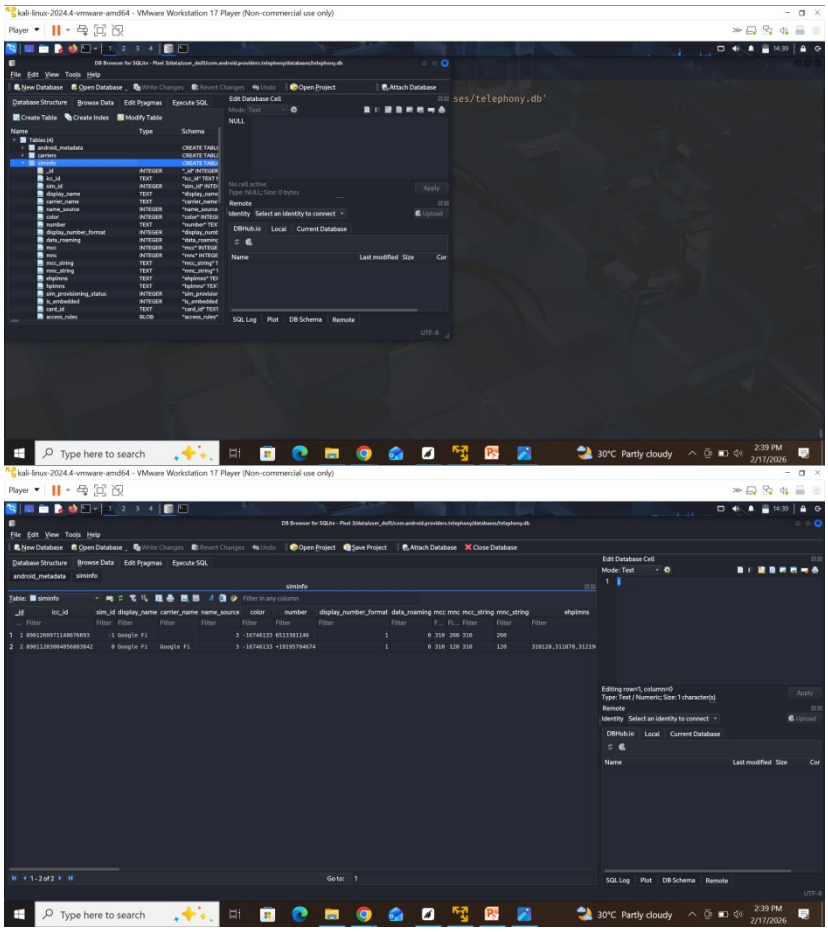


4. What is the mobile carrier/phone#?

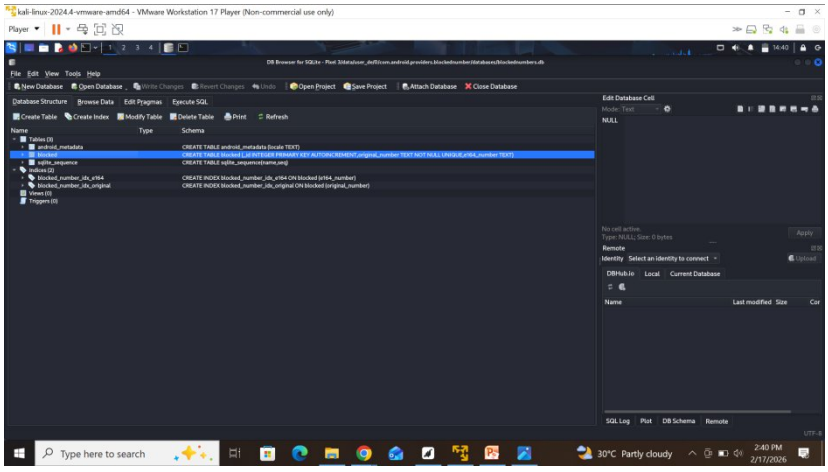


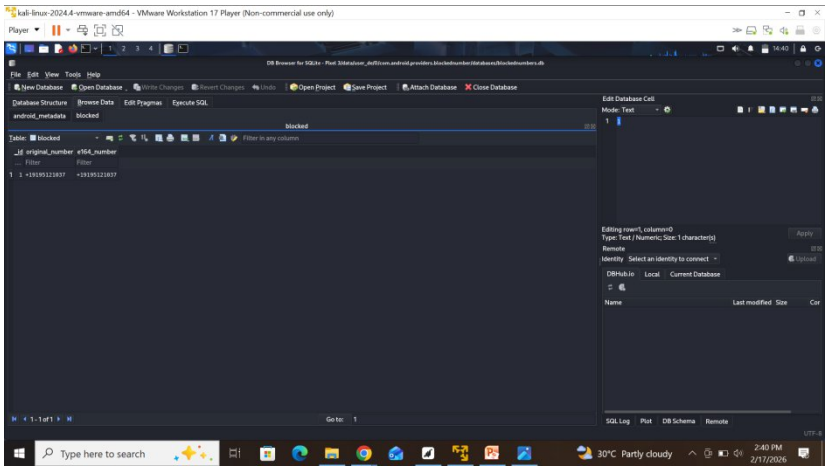
5. List tables on *telephony.db*



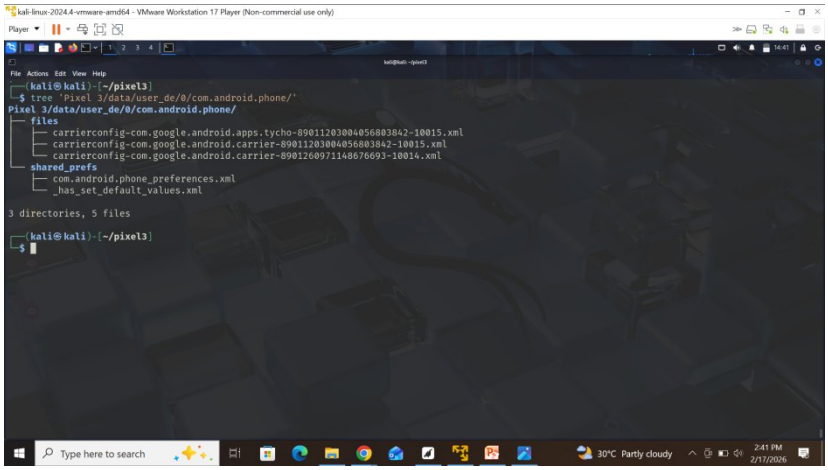


6. What is blocked phone number?

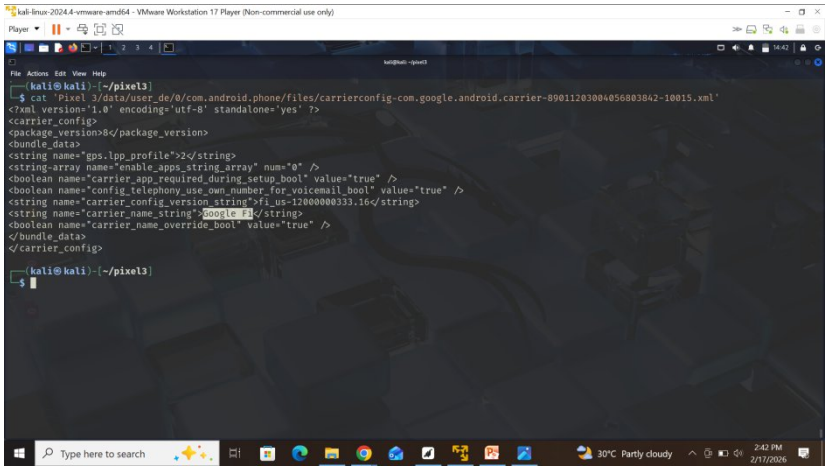


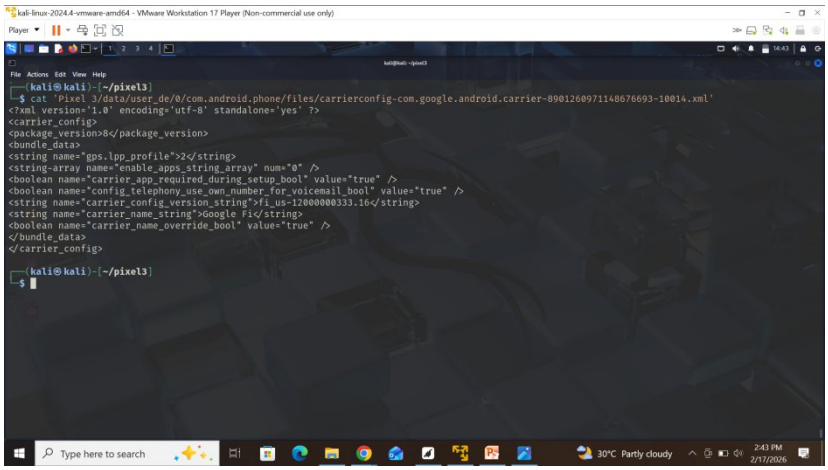


7. Check carrierconfig information to verify the carriers

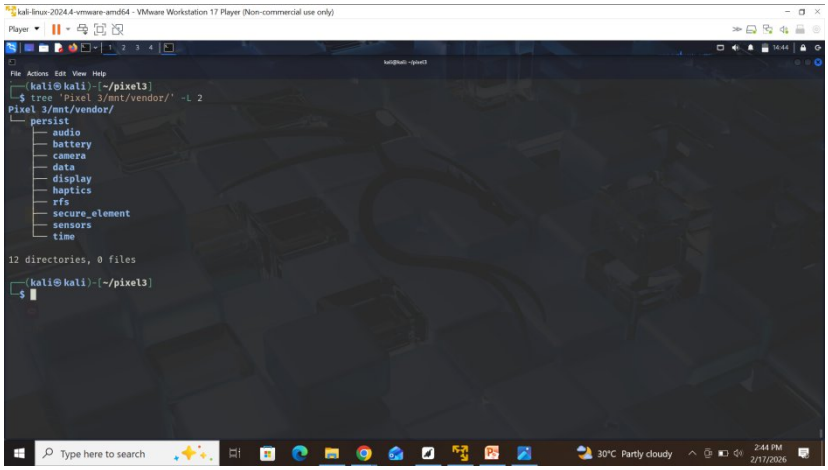


8. ICCID id

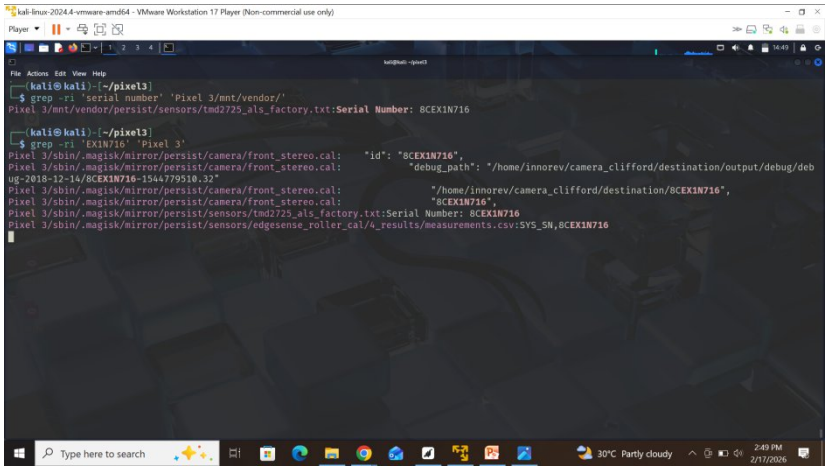




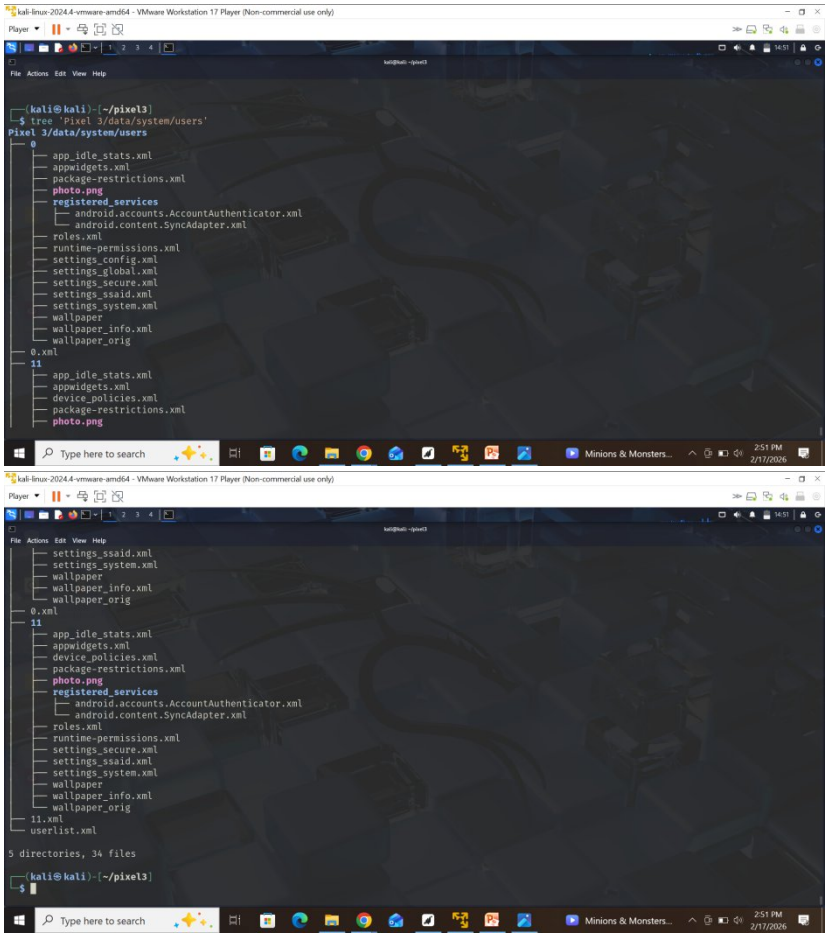
9. What is serial number of the device?



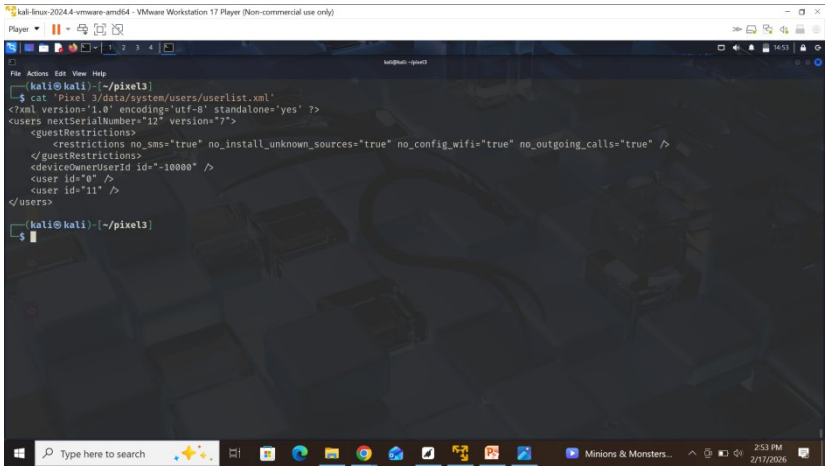
10. Search key word “serial number” under the */vendor* folder



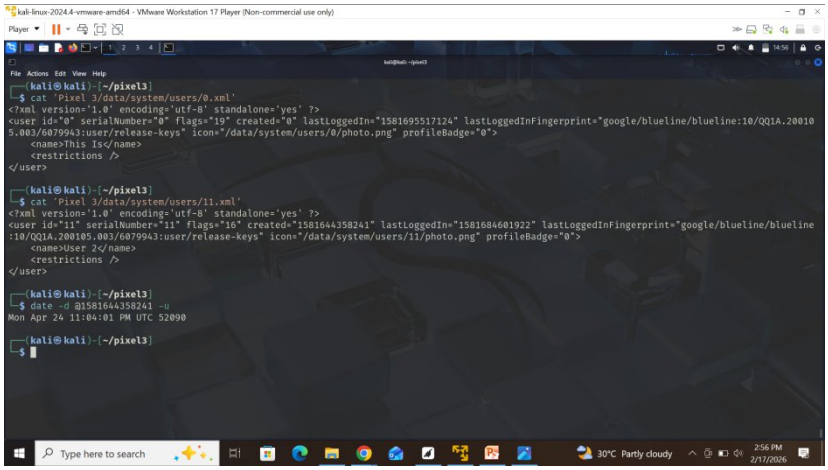
11. How many user accounts on the device? Two



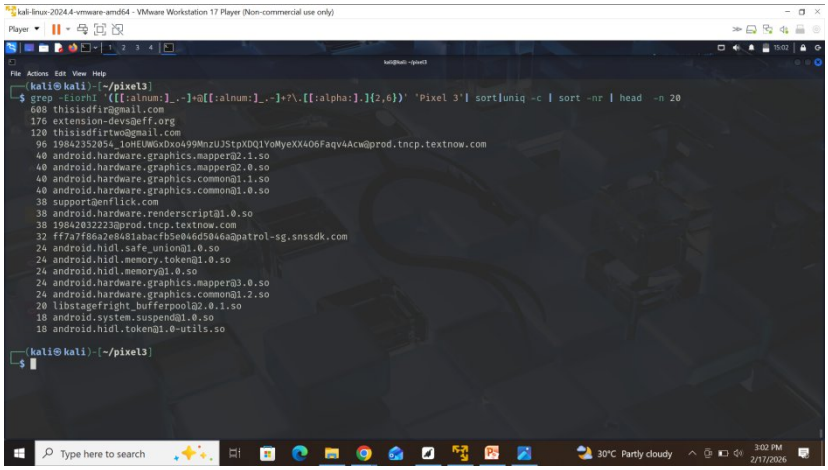
12. Who is the owner of the device?



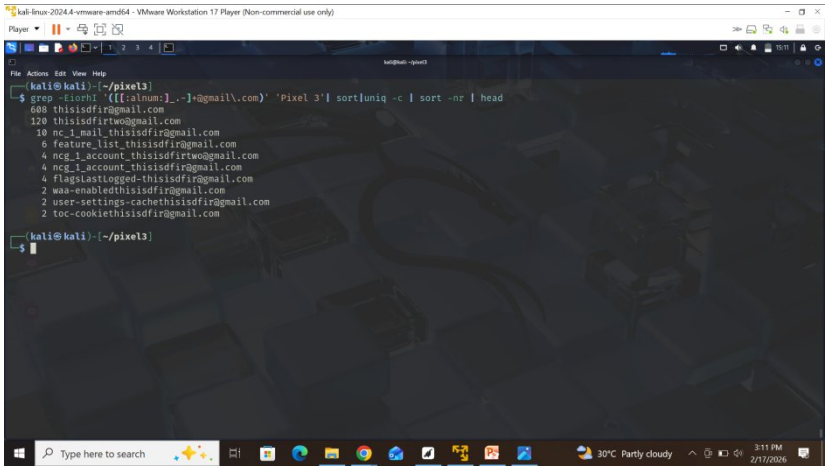
13. When did the device owner last time log in?



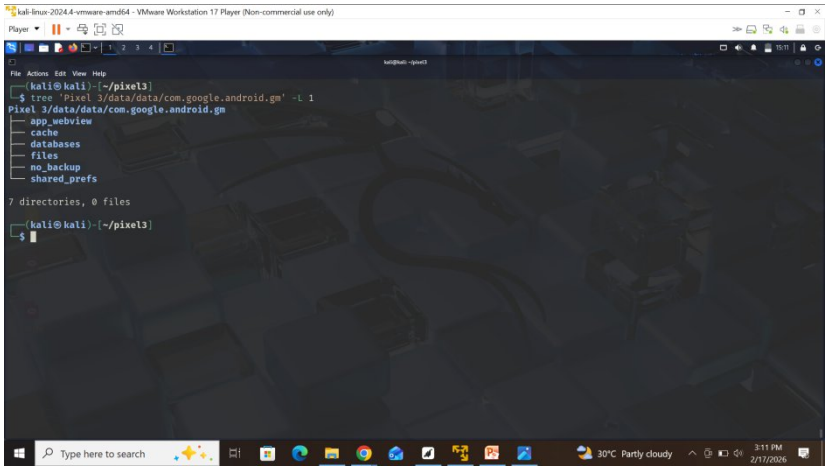
14. list top 20 email



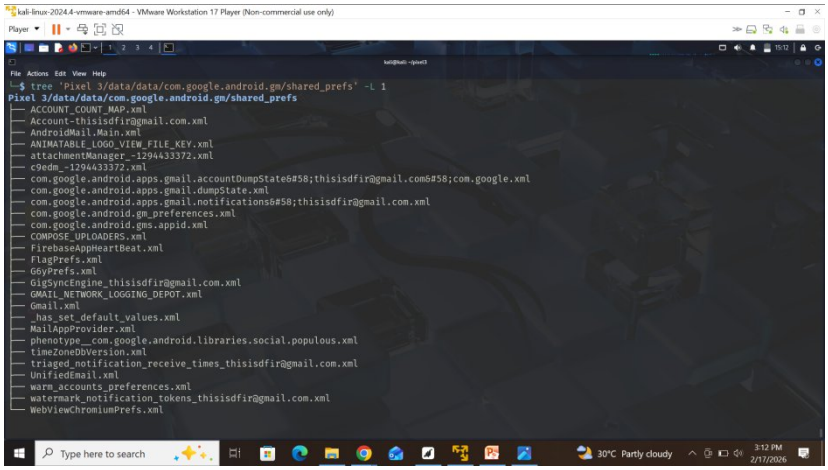
15. Find all Gmail addresses appeared in the phone?



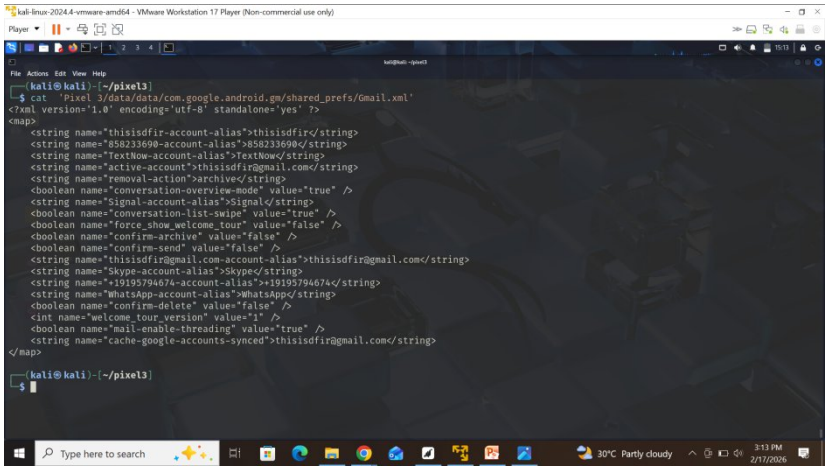
16. Which Gmail is registered for google service for Pixel 3?



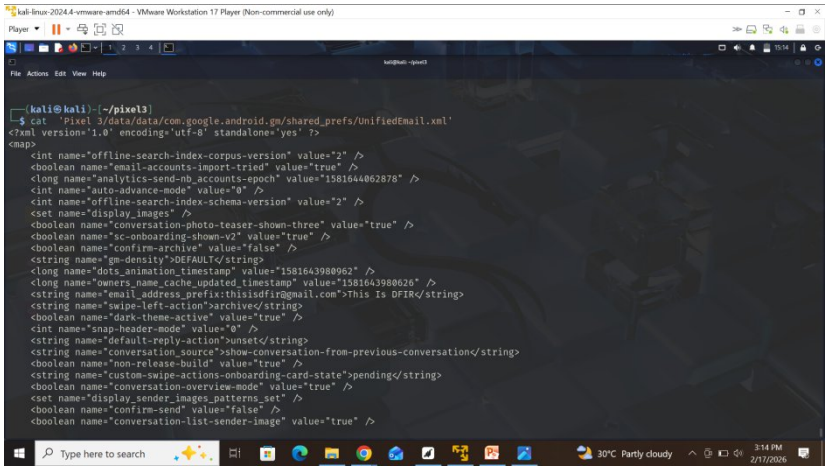
17. Show shared preference/configurations



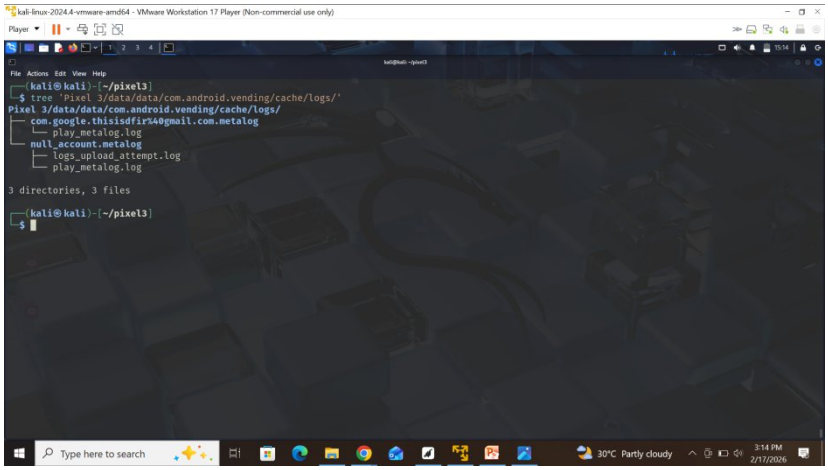
18. Check Gmail preference setting



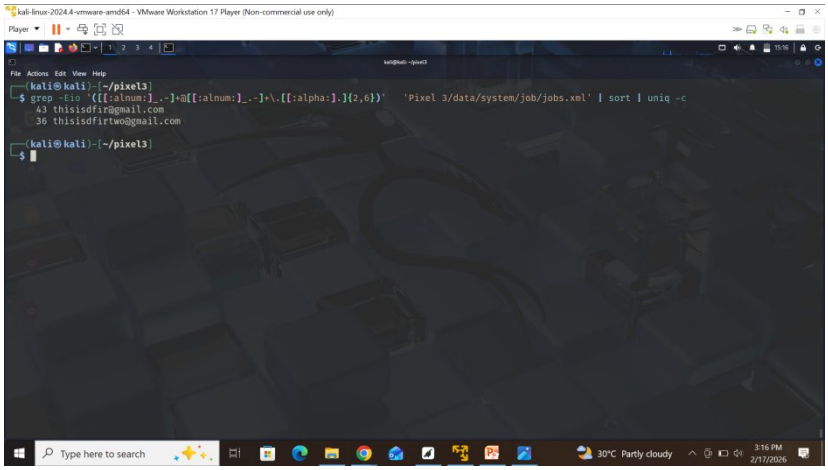
19. Gmail combines messages from multiple email accounts into one single inbox view.



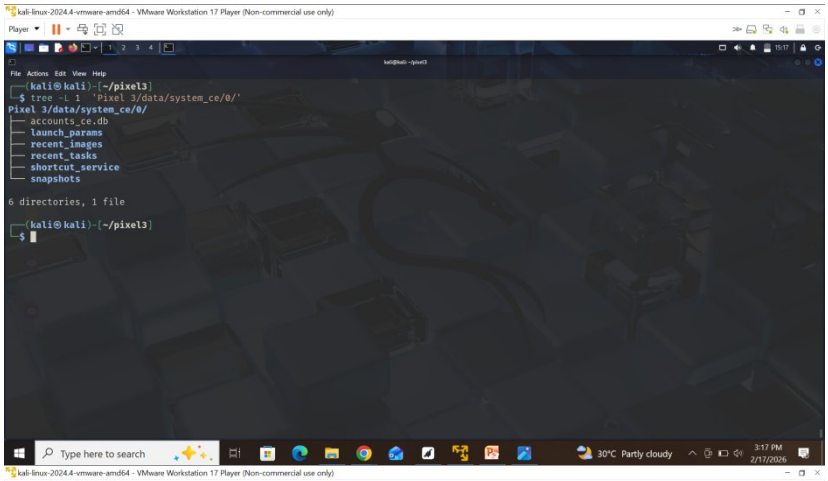
20. Which Gmail is registered in Play store?

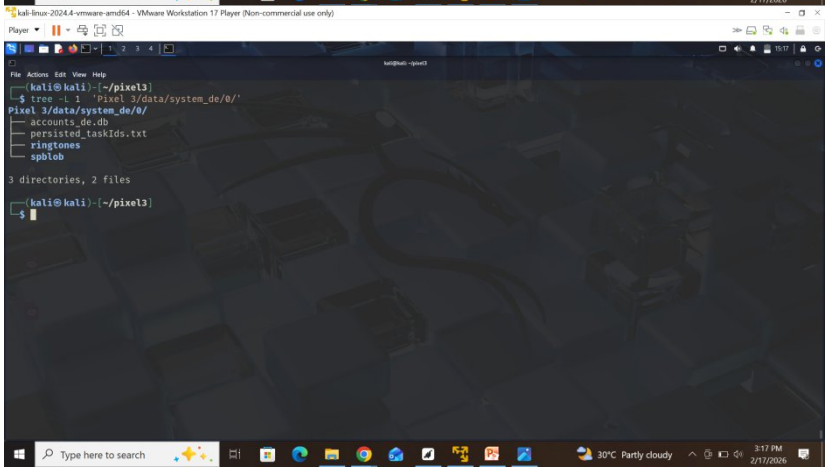


21. Find two emails that are associated with scheduled jobs

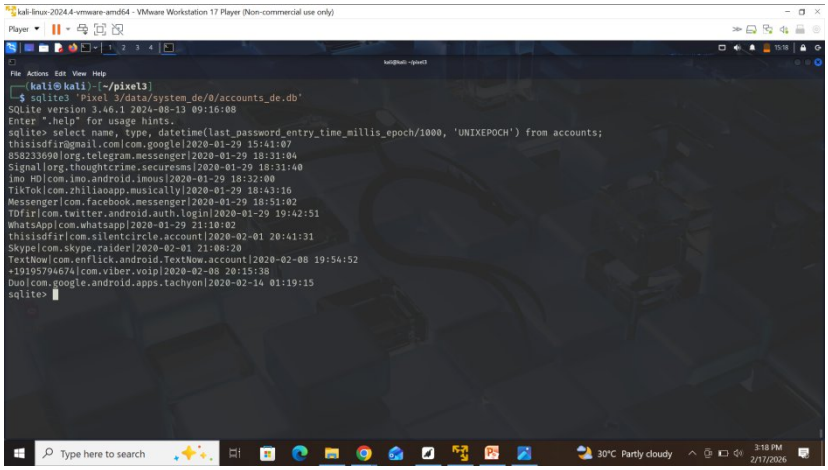


22. List possible apps username and password (account_ce.db/accounts) (account_de.db/accounts)

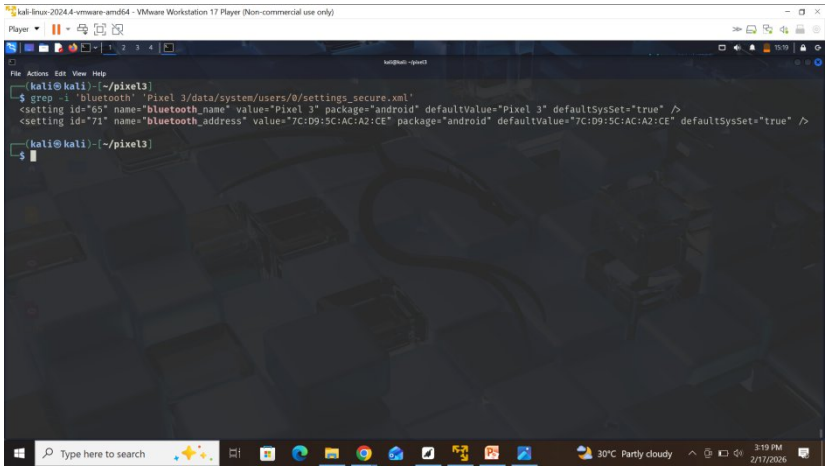




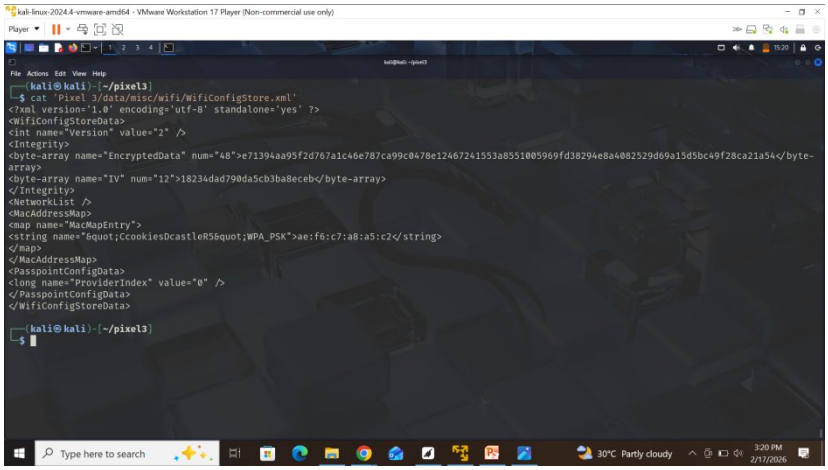
23. Convert *last password entry time millis epoch to human* readable



24. What is Bluetooth Mac address?



25. What is Wi-Fi Service Set Identifier (SSID)? (*WifiConfigStore.xml*)



Conclusion

The examination of the Pixel 3 Android 10 evidence focused on device configuration, accounts, applications, communications, and other digital artefacts to reconstruct relevant activity. Findings were assessed through source verification, timestamp analysis, and corroboration. However, conclusions are limited by the scope of the supplied logical copy and the possibility of missing data. This case study highlights the importance of evidence preservation, accurate documentation, and objective interpretation in mobile device forensics.